

Sarthak Bichhawa

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Summary

Senior Applied Scientist and org-recognized AI Champion (9+ years) building production ML, optimization, and generative-AI systems at scale—behavioral embeddings, simulation-driven testing, and autonomous LLM agents—with measurable business impact from prevented Sev1 incidents to ~\$45M/month recovered revenue.

Experience

Microsoft, Redmond, WA | *Sr. Data & Applied Scientist*

Nov 2020 – Current

Applied ML & Embeddings:

- Enabled fleet-scale similarity search, segmentation, and embedding-drift anomaly detection across **millions of accounts** by designing an **account-behavior embedding model** (autoencoder + FAISS) that compresses each Azure Storage account into a vector fingerprint.
- Caught silent upstream data regressions—including an **~88% row-collapse**—before they reached Finance by building a self-service **data-quality monitoring service** with anomaly detection and hourly table-health profiling over 47 datasets.

Optimization & Simulation:

- Built an adaptive pre-release **stress-testing program** for Azure Storage informing per-build ship/hold decisions; helped prevent **10+ Sev1/Sev2 customer incidents** by catching top-severity regressions before rollout.
- Ran **thousands of controlled simulations per build** across 20 workload types, distilling ~500 failure signals into **15 recurring failure modes** and confirming 50+ issues with reproduction steps and quantified impact.
- Prioritized high-yield test scenarios with **online learning (multi-armed bandits)** and tuned scoring thresholds on outcomes, cutting false positives **60%**; led a 2-person effort and authored the design doc + internal white paper.

AI & Agentic Systems:

- Architected an enterprise **LLM agent** that turns natural-language questions into governed live Kusto/ADX queries and PM-ready narratives, cutting customer-health time-to-insight from **days to <2 minutes** across 13 metric domains.
- Reduced multi-hour on-call incident investigations to **minutes** with an autonomous multi-agent LLM loop (planner/executor on Azure OpenAI); shipped two reusable **Model Context Protocol (MCP)** tool servers used by Copilot, VS Code, and internal agents.

Revenue & Attribution:

- Recovered and sustained **~\$45M/month in influenced partner revenue** (at <1% variance vs Finance, ~4× partner coverage) by rebuilding a 20+ component partner-attribution pipeline with **multi-partner credit** replacing last-writer-wins.

Leadership & Recognition:

- Org-recognized **AI Champion**: co-led an internal AI Day-of-Learning, mentored engineers (17 PRs reviewed, onboarded a new hire), and led a 4-team Sev-2 cross-org incident response.

Walmart Labs, Bentonville, AR | *Senior Data Scientist*

Feb 2019 – Nov 2020

- Delivered **~7% category revenue lift** across 50 stores with an end-to-end **planogram optimization** system, cutting planning cycles from weeks to hours.
- Built **product association + substitution analytics** (behavior tree + demand-transfer “universes”) to support assortment and replenishment decisions.

Integral Ad Science, New York, NY | *Data Scientist Intern*

June 2018 – Aug 2018

- Built a PySpark contextual text-classification pipeline processing **~1M URLs/day across 846 categories** (88% accuracy); improved label quality via correlation analysis.

Mu Sigma Business Solutions, Bengaluru, India | *Data Scientist*

Oct 2015 – July 2017

- Estimated **customer lifetime value (CLTV)** with probabilistic/regression models; improved competitive reporting by unifying product IDs via fuzzy matching.

Education

University of Illinois at Chicago (UIC), Chicago, IL | *MS-Business Analytics*

Aug 2017 – Dec 2018

SGSITS, Indore, India | *B.E. in Computer Engineering*

Aug 2011 – July 2015

Skills

- Analytical Techniques:** Embeddings & autoencoders (FAISS), optimization & simulation, online learning (multi-armed bandits), anomaly & drift detection, generative & agentic AI (RAG, MCP), statistical modeling/validation, classification (NLP)
- Analytical Languages & Tools:** Python, SQL, Azure Data Explorer (Kusto), PySpark, R, FastAPI, Power BI, Tableau, Azure DevOps, Git
- Functional Areas:** Production ML & AI systems, supply/demand optimization, cloud telemetry & reliability, partner/revenue attribution, customer & retail analytics